Cassava 6 — Results and Findings

Oct 26, 2026 · Anas Alsirafy

The best cassava system reaches 90.32 ± 0.78% (5-fold), within about a point of the 2020 Kaggle winner's 91.3% on its hidden test set. Nearly all of the gain over the 67% baseline came from adapting the network to field images, not from the classifier on top.

Final results

System	Accuracy (%)	Protocol	Caveat
Exp 11: EfficientNet-B0 + ConvNeXt-Base specialist, per-class weighted	90.32 ± 0.78	5-fold StratifiedKFold	Specialist folds differ from evaluation folds: optimistic
Exp 10: Noisy-Student EfficientNet-B0, MixUp/CutMix, focal loss	87.88 ± 0.90	Same	Not group-aware; check whether the held-out fold also drove callbacks
Exp 1: frozen MobileNetV2 baseline	67.32 ± 0.96	Same	—
Always guess CMD	61.5	—	Reference floor

Findings

- **Fine-tuning and resolution matter most.** Fine-tuning added 5 points (Exp 3); moving from 224 to 384 pixels added 6 (Exp 5). Augmentation on a frozen backbone added nothing (Exp 2).
- **The minority classes are the battleground.** CMD is easy; the other four classes are where focal loss, MixUp/CutMix and the specialist earned their 7–10 points.
- **Edge deployment is feasible.** The full ensemble runs at 65 ms/image on a Jetson Xavier NX and 254 ms on a Raspberry Pi 4.

Cassava in the cross-dataset study (grouped folds)

System	Accuracy (%)	Balanced acc. (%)	Macro-F1
ConvNeXt-Base, fine-tuned, 384×384 (seed 17)	86.32	76.7	0.762
ConvNeXt-Base, frozen, 384×384 (seed 17)	81.88	—	—
Frozen MobileNetV2 + EfficientNet-B0 + DenseNet121, logistic regression	70.61 ± 0.16	53.6	0.534

- Fine-tuned beats frozen small backbones by 15.7 points (95% CI 14.5–17.0, $p < 10^{-5}$).
- The frozen ConvNeXt-Base closes about three-quarters of that gap; fine-tuning adds the last 4.4 points (95% CI 2.1–6.7, $p = 0.006$).
- With frozen small backbones, CMD reaches F1 0.86 but the other four classes only 0.38–0.50 — just 9.1 points above always guessing CMD.
- Cost: a fine-tuned fold took about 262 minutes; the frozen-backbone control 170 minutes; the logistic regression 78 seconds.

Against published results

Work	Method	Protocol	Accuracy (%)
Riaz et al.	EfficientNet-B3	Own test split	83.0
Fang et al.	ConvNeXt-T, fine-tuned	80/20	88.9
Fang et al.	VGG-13 BN, fine-tuned	80/20	89.0
Kaggle 2020 winner	4-model ensemble (ResNeXt50, ViT-B/16, EfficientNet-B4 Noisy Student, CropNet MobileNetV3)	Hidden test set	91.3
This work, Exp 11	Weighted 2-model ensemble	5-fold	90.3
This work, cross-dataset	ConvNeXt-Base, fine-tuned	Grouped 5-fold, 1 seed	86.3

Full references are in Paper 2, Table IV.